BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 088 · Logic Gates: AND, OR and NOT

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What must be true before binary search is used?

Q2. What does software copyright restrict?
A. Using any keyboard
B. Learning to code
C. Saving your own work
D. Copying without permission

Q3. “Proprietary software always gives every user unrestricted source-code modification rights.” Correct this. Explain why.

Q4. A ticket program follows the language's rules but adds 5 instead of multiplying by 5. “Logic errors always stop translation.” Correct this. How should the error be found?

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I Do · Read new knowledge

Key vocabulary

AND — Output 1 only when both inputs are 1.

OR — Output 1 when either or both inputs are 1.

NOT — Reverses one Boolean input.

Gate — Electronic circuit implementing a Boolean rule.

Inversion bubble — Small circle on a symbol indicating logical inversion.

Boolean — Logic with values 0 and 1 (false and true).

New learning

Binary logic: Logic gates work with Boolean input and output values 0 and 1.

AND gate: AND outputs 1 only if both inputs are 1. Its symbol has a flat input edge and rounded output edge.

OR gate: OR outputs 1 if either or both inputs are 1; it outputs 0 only for 0,0.

NOT gate: NOT has one input and outputs the opposite value; its symbol has a small inversion bubble.

Worked example

An alarm sounds only if a door is open (A=1) AND the system is armed (B=1). Explain the outputs for A=1,B=1 and A=1,B=0. [4 marks]

1. This is an AND rule because both conditions must be true [1].

2. With 1,1 the output is 1 [1], so the alarm sounds.

3. With 1,0 the output is 0 [1] because both are not true [1].

4. Show each input pair alongside its output before revealing.

We Do · Practise with guidance

Guided 1

A buzzer sounds if either A or B is on. Give OR outputs for A,B = 00,01,10,11. [4 marks]

Worked method: 0,1,1,1 in the given order, one mark for each correct output.

Guided 2

A safety indicator is NOT A. Give output for A=0 and A=1. [2 marks]

Worked method: For A=0 output 1 [1]; for A=1 output 0 [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Guided 3

A fan needs power A and permission B. Explain why AND is suitable and state outputs for 1,0 and 1,1. [4 marks]

Worked method: Both inputs must be high [1], so AND is suitable [1]. 1 AND 0=0 [1]; 1 AND 1=1 [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the Boolean test so both security checks must be true.

```
signedIn = true
hasPermission = true
if ... then
    print("Allow")
endif
```

write

Question 1

State the output of an AND gate given A=1 and B=0. [1 mark] Source: Original OCR J277-style practice question.

Question 2

Give OR outputs for input pairs 00, 01, 10 and 11 in that order. [4 marks] Source: Original OCR J277-style practice question.

Question 3

Explain AND, OR and NOT with TWO different inputs or examples for each. [6 marks] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

An OR gate outputs 1 when both its inputs are 1.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The list must be sorted. This lets the search safely rule out half the remaining items.

Q2. D — Copying without permission

Q3. Source code is normally controlled by its owner and users receive specified usage rights.

Q4. A logic error gives the wrong result even when the code follows the language's rules. Fix syntax errors first, then trace and test the logic.

I Do / We Do pages · Guided practice

Q1. 0,1,1,1 in the given order, one mark for each correct output.

Q2. For A=0 output 1 [1]; for A=1 output 0 [1].

Q3. Both inputs must be high [1], so AND is suitable [1]. 1 AND 0=0 [1]; 1 AND 1=1 [1].

You Do pages · Code First

```
signedIn = true
hasPermission = true
if signedIn AND hasPermission then
    print("Allow")
endif
```

Expected result: Allow

You Do pages · Independent questions

Q1. 0 [1].

Q2. 0 [1], 1 [1], 1 [1], 1 [1].

Q3. AND 1,1→1 [1] and 1,0→0 [1]. OR 0,1→1 [1] and 0,0→0 [1]. NOT input 0→1 [1] and input 1→0 [1].

Exit Ticket · Model answer

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: